

Assignment: Layer 3 Switch Inter-VLAN Routing

Simple student-level explanation with Cisco CLI commands

Aim of the Assignment

The aim of this practical is to learn how a Layer 3 switch can route traffic between different VLANs. In this task, I create three VLANs, give each VLAN an SVI IP address, enable IP routing, test communication between PCs, and finally change one VLAN to a different subnet mask.

Basic Topology

Device	Purpose
Layer 3 Switch	Creates VLANs and performs inter-VLAN routing
PC 1	VLAN 10 - Sales
PC 2	VLAN 20 - Support
Optional PC 3	VLAN 30 - Marketing

Question 1 - Create Three VLANs and Assign Ports

Question: On a Cisco Layer 3 switch, create VLAN 10 for Sales, VLAN 20 for Support, and VLAN 30 for Marketing. Assign at least one interface to each VLAN.

Answer: First, I create the three VLANs. Then I put one access port in each VLAN.

```
enable
configure terminal

vlan 10
 name Sales
exit

vlan 20
 name Support
exit

vlan 30
 name Marketing
exit

interface fastethernet 0/1
 switchport mode access
 switchport access vlan 10
exit

interface fastethernet 0/2
 switchport mode access
 switchport access vlan 20
exit

interface fastethernet 0/3
 switchport mode access
 switchport access vlan 30
exit
```

```
end
write memory
```

Verification: show vlan brief should show VLAN 10, 20 and 30 as active, with the correct ports.

Question 2 - Configure SVIs

Question: Configure a Switch Virtual Interface (SVI) for each VLAN with a unique IP address and make sure each SVI is up/up.

Answer: I use VLAN interfaces as the default gateways for the three VLANs.

```
enable
configure terminal

interface vlan 10
ip address 192.168.10.1 255.255.255.0
no shutdown
exit

interface vlan 20
ip address 192.168.20.1 255.255.255.0
no shutdown
exit

interface vlan 30
ip address 192.168.30.1 255.255.255.0
no shutdown
exit

end
write memory
```

Note: The SVI IP is the default gateway for the PCs in that VLAN.

Verification: Run show ip interface brief. The VLAN interfaces should show an up/up state when the VLAN is active and at least one port in that VLAN is up.

Question 3 - Enable IP Routing

Question: Enable routing on the Layer 3 switch so devices in different VLANs can communicate without an external router.

Answer: The command ip routing enables the Layer 3 switch to route between the VLAN networks.

```
enable
configure terminal
ip routing
end
write memory
```

Question 4 - Verify Inter-VLAN Routing

Question: Connect two PCs, one in VLAN 10 and one in VLAN 20, and test communication by pinging between them.

Answer: I assign one PC to VLAN 10 and one PC to VLAN 20. Their default gateways are the SVI addresses of their VLANs. After enabling IP routing, the two PCs should be able to ping each other.

PC	VLAN	IP Address	Subnet Mask	Default Gateway
PC1	10	192.168.10.10	255.255.255.0	192.168.10.1
PC2	20	192.168.20.10	255.255.255.0	192.168.20.1

```
PC1> ping 192.168.20.10
```

```
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
```

Result: The ping is successful, which shows that the Layer 3 switch is routing traffic from VLAN 10 to VLAN 20.

Question 5 - Change VLAN 30 to /25

Question: Refactor VLAN 30 so it uses a different subnet mask, for example /25 instead of /24. Explain what was changed and how this affects communication with the other VLANs.

Answer: A /25 mask gives VLAN 30 a smaller subnet. The original /24 network has more host addresses, while /25 divides the /24 space into two smaller subnets. I keep the VLAN 30 SVI inside the new /25 network and make sure the VLAN 30 PC also uses the same /25 mask.

```
interface vlan 30
ip address 192.168.30.1 255.255.255.128
no shutdown
exit
```

Example VLAN 30 PC: IP = 192.168.30.10, Mask = 255.255.255.128, Gateway = 192.168.30.1.

Effect: VLAN 30 can still communicate with VLAN 10 and VLAN 20 through the Layer 3 switch because routing is enabled. However, the VLAN 30 PC and SVI must be in the same /25 subnet. If the IP/mask settings do not match, communication can fail.

Useful Verification Commands

```
show vlan brief
show ip interface brief
show running-config
show ip route
ping <destination-ip>
```

Conclusion

In this practical, I learned that a Layer 3 switch can do the job of both a VLAN switch and a router. VLANs separate the network, SVIs provide gateways for each VLAN, and the `ip routing` command allows communication between different VLANs. I also learned that changing the subnet mask changes the size of the IP network, so the SVI and PC addressing must match.

Submission note: Capture screenshots of `show vlan brief`, `show ip interface brief`, and the successful PC ping. These screenshots can be added to the final submission as proof of the practical.